When constant g is not enough

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Recognize the limits of the near-surface mgh model.

SUCCESS CRITERIA - I can...

1. Compare altitude with planetary radius.
2. Explain that gravitational field weakens with center distance.
3. Choose a model appropriate to the height interval.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 kg cart starts at rest. A 19.5 N pull and 7 N resistance act over 3 m. Find final speed.

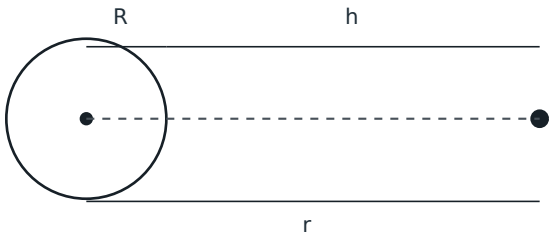
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta U \approx mg\,\Delta h$; $g = GM/r^2$.
The first relation needs nearly constant g. For large changes in r, the field changes; one fixed surface value is insufficient.

Worked reasoning: A local model treats g as constant over small height changes. At a center distance twice a planet's radius, however, g is only one quarter of its surface value.

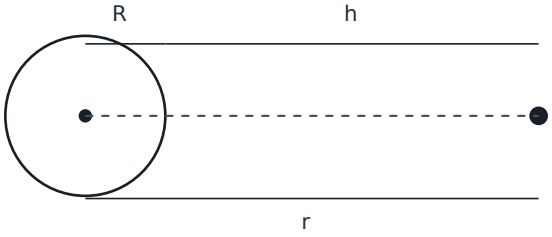
Your turn: Sketch a planet and mark center distances R and 2 R. Explain why one surface g cannot describe the entire trip exactly.



Marzano check - circle: ① ② ③

03 Read and interpret the model

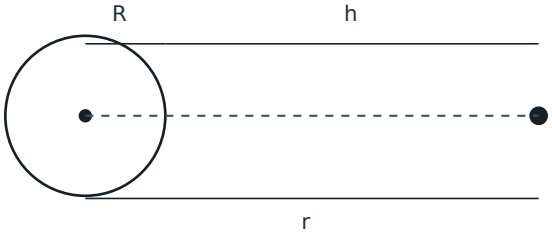
Compare lifting a book 1 m with moving a satellite one planetary radius above the surface. Which better fits constant g and why?



Marzano check - circle: ① ② ③

04 Apply the physics

A gravity simulation uses $GM = 100\ m^3/s^2$. A 2 kg object moves from center distance 10 m to 20 m. Label both distances. Calculate $\Delta U = GMm(1/r_i - 1/r_f)$. Compare with mgh using the starting field $g = 1\ m/s^2$ and h = 10 m.



Marzano check - circle: ① ② ③

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Continue down each column.

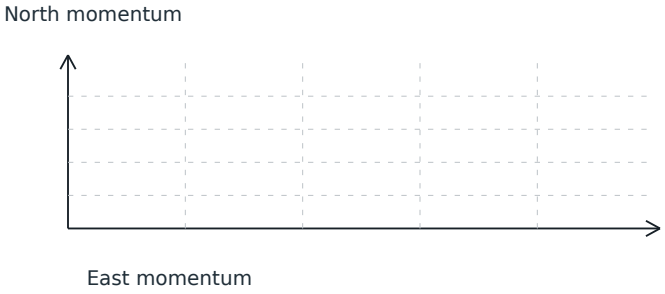
05 Explain your reasoning

Why is altitude h not interchangeable with center-to-center r in the universal expression?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

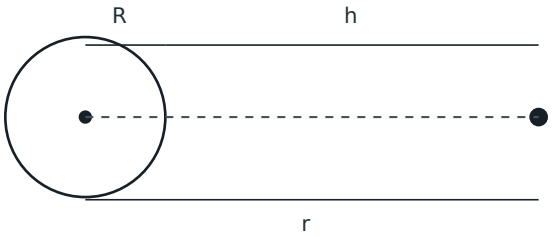
Total momentum initially is (24, 0) kg·m/s. One fragment leaves with (4, 6) kg·m/s. Find the other momentum vector and magnitude.
Label momentum components and their sum with units.



Marzano check - circle: 1 2 3

07 Exit check

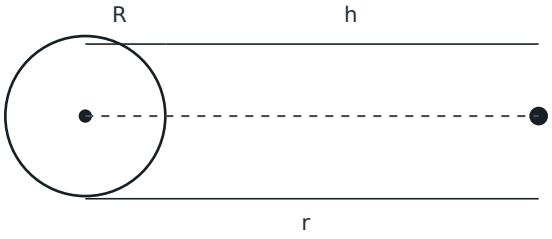
Using the same simulated planet, compare gravity at r=10 and 20 m. Connect the result to the energy-model choice. Label the relevant heights or distances on the model.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A planet has GM=100 m³/s² and radius 5 m. Compare exact gravitational potential-energy increase for a 2 kg object raised to r=10 m with the surface-g approximation. Label the relevant heights or distances on the model.



Marzano check - circle: 1 2 3